Module 0: Setup & Welcome

- **Teaching Objective:** Ensure technical readiness.
- **Learning Outcome:** Participants will have a fully functioning R/RStudio environment with NIMBLE, Cytoscape, and `pgdcm` successfully installed and configured.

Slide 1: Title & Welcome

Module 1: The "Why" and The "Wow" (run_pgdcm_auto)

- **Teaching Objective:** Introduce the pgdcm as a unifying framework linking IRT, CDM, and SEM , and demonstrate a complete model run from start to finish.
- **Learning Outcome:** Participants will be able to articulate the conceptual purpose of psychometric modeling and successfully execute a standard automated pipeline to generate basic parameter estimates and latent skill profiles in minutes. [acknowledge prior predictive results and posterior predictive results but don't explain them - we will tell them to wait until module 4.]

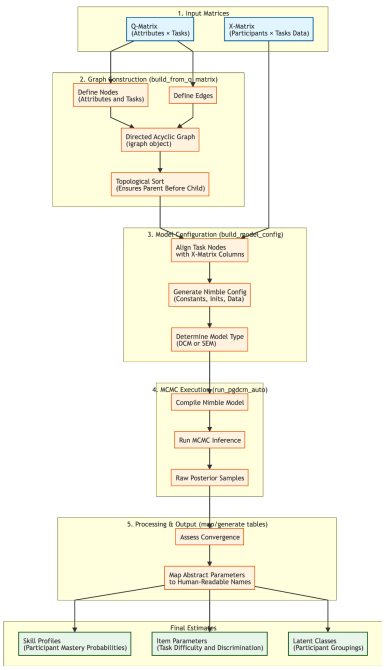
Slide 2: What are psychometric models? How are they different from other statistical models ATHUL

Slide 3: Commonly used Psychometric Models: IRT, DCMs, SEMs ATHUL

Slide 4: Represent all these psychometric models as graph models side by side ATHUL

Slide 5: PGDCM Mental Model ATHUL

Show the below flow diagram



Hands-On exercise 1: ATHUL

1. Participants will load a dataset, q-matrix and run the run_pgdcm_auto()
2. We'll spend some time looking at the skill profile matrix csv, the item parameter estimates csv as the latent class membership (clustered profiles) csv file.

Module 2: Encoding Knowledge as Bayesian Networks

- **Teaching Objective:** Teach the specification of competency and evidence models using graphical modeling syntax, grounded in Evidence-Centered Design (ECD).
- **Learning Outcome:** Participants will be able to formalize a model's structure as a directed graph object using Cytoscape and iGraph, demonstrating the ability to move beyond simple Q-matrices to map custom, hierarchical cognitive architectures.

Slide 6 - 8: Bayesian Networks 101: Explain what a node and an edge represent in a Bayesian network. Discuss how to semantically interpret the network and how the graph structure allows for efficient factorization of the likelihood. Dr. GOLDEN

Slide 9: Evidence Centred Design 101: Explain what is a proficiency model and evidence model. What's a task node and what's an attribute node. Dr. GOLDEN

Slide 10: A World beyond Q: How Q Matrix can be viewed as nodes and edges Dr. GOLDEN

Hands-on Exercise 2: ATHUL

1. Participants will load a nodes file and edges file to R and visualize them as a graph in R and save the object as a graphml object
2. Participants will load the Cytoscape template file, connect to Cytoscape via RCy3 and try to recreate the nodes and edges from the previous example using the Cytoscape UI. Fetch the graph from Cytoscape to R and save it as a graphml object.

Module 3: Under the Hood - Bayes & Estimation

- **Teaching Objective:** Explain how to translate graphical structures into statistical models and estimate parameters using MCMC methods.
- **Learning Outcome:** Participants will be able to implement bayesian psychometric models using NIMBLE's syntax, manipulate MCMC configuration settings, customize priors, and infer student latent skill profiles

Slide 11: Bayes 101: What is Bayesian modelling, and how does it compare to the frequentist approach?. Discuss the pros and cons of each. Explain the Bayes Formula: Data + Likelihood + Prior Dr. GOLDEN

Slide 12: Why NIMBLE? Compare available tools (JAGS/NIMBLE/Stan). Explain why NIMBLE is the backend for `pgdcm`. Dr. GOLDEN

Slide 13: NIMBLE Syntax. Show a simple 2PL IRT model in NIMBLE code purely to explain the concepts (clarify that we won't be running this specific code at this stage). ATHUL

Slide 14: Breaking apart run_pgdcm_auto: Show 12 lines of code that implements the entire workflow and give a high level overview. ATHUL

Hands on exercise 3:ATHUL

1. Participants will use the bayesian network developed during previous hand-on exercise to pass it to pgdcm, change the configuration of MCMC config and run the MCMC estimation.

Module 4: Validation & Diagnostics

- **Teaching Objective:** Demonstrate how to evaluate model quality and robustness using simulation-based experiments.
- **Learning Outcome:** Participants will be able to diagnose MCMC convergence and evaluate overall model fit using Widely Applicable Information Criterion (WAIC), prior predictive checks, and posterior predictive checks.

Slide 15: Predictive Checks. Explain Prior Predictive Checks (ensuring the model doesn't over-constrain the data space) and Posterior Predictive Checks (regenerating the observed X-Matrix). Dr. GOLDEN

Slide 16: Did it Converge? How to read trace plots, density plots, and the R^ (R-hat) statistic and the RMG convergence criteria. Dr. GOLDEN

Slide 16: Relative Model fit: Introduce WAIC, lppd, and pWAIC for comparing different Bayesian models.Dr. GOLDEN

Module 5: Scoring

- **Teaching Objective:** Demonstrate how to pass the item parameter estimates from a previous run as strong priors to `pgdcm` and use the model as an operational scoring engine.
- **Learning Outcome:** Participants will be able to lock item parameters using historical estimates and execute a modified `pgdcm` run specifically to infer latent skill profiles for a new batch of students.

Hands on Exercise 4: ATHUL

1. Participants will take the item parameter output CSV from hands on exercise 3. Map these estimates as strong priors for a brand new dataset of student responses and do estimation using run_pgdcm_auto or the earlier script they developed in module 3.